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# Project-based learning as a pathway to writing autonomy and responsible AI use in university EFL learners

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**Introduction:** The rapid adoption of artificial intelligence (AI) tools such as ChatGPT for academic writing has raised pedagogical concerns regarding learners' writing autonomy, self-regulation, and ethical use of AI in higher education. This study investigates changes in writing performance and reported AI usage patterns following the implementation of Project-Based Learning (PBL) among university-level EFL learners.

**Methods:** A quasi-experimental one-group pretest -posttest design was implemented with A2-level learners from Universidad Central del Ecuador. Participants completed writing tasks modeled after the Cambridge A2 Key exam to assess organization, grammar, vocabulary, and coherence before and after a short-term, one-week PBL intervention conducted under restricted AI conditions. Surveys measured learners' writing processes, self-regulation behaviors, and self-reported ChatGPT usage, while instructor surveys captured perceptions of collaboration and motivation.

**Results:** Results indicated a statistically significant improvement in writing performance following the brief intervention [ $t(43) = 4.15$ ,  $p < .001$ ], representing a change in paired sample scores between the pretest and posttest. Survey responses indicated frequent use of AI at the beginning of the study and more selective interaction during the intervention period; however, these patterns were based on self-reported data. Teachers described learners as highly motivated and collaborative during PBL activities, although challenges remained in task division and group problem-solving.

**Discussion:** Considering the quasi-experimental design, the short duration of the intervention, and the absence of a control group, the results should be interpreted with caution and do not constitute definitive evidence of the causal effect of PBL. Nevertheless, the findings suggest the potential impact of instructional design on writing development and the integration of AI tools in EFL teaching contexts, highlighting the need for more comprehensive and controlled research.

### KEYWORDS

AI literacy, ChatGPT in education, EFL writing, higher education, project-based learning, responsible AI use, writing autonomy

# 1 Introduction

Project-Based Learning (PBL) has gained increasing attention in language education as a learner-centered instructional approach associated with autonomy, collaboration, and meaningful engagement with authentic tasks. In EFL contexts, PBL has been linked to reflective thinking, self-directed learning, and learners' ownership of their linguistic development (Kim, 2024; Simatupang and Yusuf, 2024). Research suggests that when learners participate in real-world, collaborative activities, they often demonstrate greater persistence, motivation, and responsibility for their learning outcomes—factors commonly associated with writing autonomy.

At the same time, the rapid expansion of artificial intelligence (AI) tools such as ChatGPT has transformed writing processes in higher education. ChatGPT provides personalized interaction, immediate feedback, and linguistic support that may strengthen vocabulary development, grammatical accuracy, and idea generation (Jo, 2024; Pikhart et al., 2024). However, these affordances have also generated concerns regarding academic integrity, potential overreliance on AI-generated content, reduced cognitive engagement, and fewer opportunities for learners to practice independent writing processes (Tai et al., 2023). While AI tools can function as scaffolds, unstructured or uncritical use may limit engagement in planning, monitoring, and revision. Thus, increasing attention has been directed toward promoting responsible and reflective AI use rather than discouraging the technology outright.

This study takes a more pedagogical than technical perspective on AI. It does not differentiate between specific versions of the ChatGPT model (e.g., GPT-3.5 vs. GPT-4), as it focuses on observable classroom practices and usage patterns established by learners, rather than comparative model performance. Since learners accessed ChatGPT according to their usual habits, variations in model versions were not controlled, and they are acknowledged as a contextual limitation rather than a variable subject to research.

In the Ecuadorian higher education context, the integration of AI into academic practices remains relatively recent. Although university instructors frequently use ChatGPT, only a minority have received formal preparation in pedagogical or ethical AI implementation (Espinoza et al., s. f.). At the regional level in Latin America, EFL teaching continues to be challenged by structural issues, such as class size, inconsistent prior preparation, and limited exposure to more extensive academic writing tasks. These contextual challenges make it harder to develop sustainable writing skills and independent learning habits. A research found that many learners perceive writing as time-intensive and frustrating, underscoring the importance of engaging instructional approaches such as PBL to increase relevance and motivation (Alvarado et al., 2023).

Low national proficiency levels further emphasize this need. According to the 2023 EF English Proficiency Index, Ecuador ranks among the lowest in Latin America, with many learners entering higher education below the CEFR-recommended B2 level for university academic work (Council of Europe, 2020). The Academic Language Institute at Universidad Central del Ecuador, which has promoted communicative methodologies through partnerships such as Cambridge University Press

(Universidad Central del Ecuador, 2018), provides a relevant setting to explore innovative approaches that combine PBL, writing development, and responsible AI integration.

In this context, PBL may offer a structured environment in which learners engage in collaborative planning, drafting, and revision processes that resemble authentic communicative practices. Its emphasis on problem-solving and shared accountability aligns with metacognitive and self-regulatory behaviors associated with writing development. Nevertheless, the extent to which short-term PBL interventions may influence measurable writing performance or AI usage patterns has not yet been empirically established, especially in contexts in which access to AI and educational practices are evolving at the same time.

Importantly, the present study does not attempt to directly quantify internal cognitive constructs like decision-making ability or autonomy. Instead, it investigates behavioral indicators, such as changes in written grades, self-reported AI usage behaviors, and performance in AI-restricted contexts. Therefore, any interpretations regarding autonomy are derived from observable behaviors, rather than directly tested psychological processes.

## 1.1 Research objectives

### 1.1.1 General objective

To examine whether participation in a short-term Project-Based Learning (PBL) intervention is associated with changes in writing performance, behavioral indicators of writing autonomy, and self-reported AI usage among university EFL learners.

### 1.1.2 Specific objectives

1. To measure changes in learners' writing proficiency before and after a PBL intervention using A2 Key pre- and post-test scores.
2. To analyze behavioral indicators associated with writing autonomy including planning behaviors, self-checking strategies, and writing performance under AI-restricted conditions.
3. To identify changes in learners' self-reported patterns of ChatGPT use, including frequency, purpose, and perceived dependence, following the PBL intervention.
4. To evaluate instructors' perceptions of learners' collaboration, task division, and problem-solving during PBL activities.

### 1.1.3 Research questions

Guided by this rationale, the present study examines Project-Based Learning within a quasi-experimental one-group pretest-posttest design involving A2-level university learners completing a one-week PBL intervention.

Taking into account the absence of a control group and changes in AI conditions between tests (AI was permitted in the

pre-test and restricted in the post-test), the results are interpreted as changes observed over time and not as definitive causal effects of PBL only.

The study addresses the following *research questions*:

1. How is participation in a short-term PBL intervention associated with changes in learners' writing proficiency?
2. How do learners' self-reported patterns of ChatGPT use change following the intervention?
3. In what ways are behavioral indicators of writing autonomy—such as planning, self-monitoring, and confidence when writing without AI tools—reflected in post-test performance? How do instructors describe learners' collaboration, task division, and problem-solving during PBL activities?

This research contributes significantly to existing debates on AI literacy and pedagogical innovation in EFL higher education, as it provides exploratory empirical data from the Latin American setting. Instead of making definitive causal statements, the study offers conclusions specific to the context in which it was developed, providing a basis for future controlled or long-term research on the relationship between PBL, writing development, and the responsible use of AI.

## 2 Literature review

### 2.1 Project-based learning (PBL) in EFL writing development

PBL represents an instructional strategy designed to foster learner engagement through collaborative tasks and authentic problem-solving situations. It promotes analytical reasoning, meaningful communication, and responsibility for learning outcomes. In EFL writing contexts, PBL shifts instruction from isolated grammar exercises to purposeful writing tasks embedded in real-world scenarios. Instead of approaching writing as a product, PBL focuses on it as a process that requires planning, negotiation, revision, and reflection, which are skills directly related to autonomous language development.

#### 2.1.1 Critical thinking and problem-solving

Critical thinking and problem-solving are central to project-based instruction. Structured models reinforce learners' interpretive, evaluative and explanatory abilities (Rohani et al., 2021; Sung and Hwang, 2013) explain that PBL supports these skills by engaging learners in solving authentic problems collaboratively.

Problem-solving approaches are particularly relevant in writing instruction, as they require learners to analyze information, construct arguments, and justify positions. Research found that reflective writing during inquiry-based tasks enables learners to synthesize knowledge and develop more nuanced arguments (Hornstein et al., 2025). Therefore, in contexts of EFL writing, PBL enhances cognitive engagement by requiring learners to generate original ideas rather than reproduce pre-structured responses, which is especially relevant when AI tools are available.

#### 2.1.2 Self-directed learning and engagement

PBL encourages learners to set goals, manage time, and monitor progress, thereby fostering self-directed learning (Bagheri et al., 2013). Reflection and self-assessment embedded in PBL have been associated with deeper cognitive engagement and improved language outcomes (Astawa et al., 2017).

Behavioral and motivational engagement also increase in project-based environments. Simatupang and Yusuf report that PBL enhances participation, persistence, and social interaction (Simatupang and Yusuf, 2024). Maulina (2024) further notes that meaningful projects promote intrinsic motivation.

All these findings together place PBL not only as a teaching method, but as an approach that combines cognitive engagement, collaboration, and self-regulation, which are key aspects for autonomy in writing.

### 2.2 Writing autonomy in higher education contexts

Autonomy in second language writing refers to learners' capacity to take control of their learning processes (Benson, 2011). In writing, this includes decision-making about planning, drafting, revising, and evaluating texts.

Zimmerman (2000) conceptualizes self-regulated learning as a cyclical process involving planning, monitoring, and self-evaluation. In writing contexts, self-regulation leads to autonomy, since learners turn abstract control into concrete strategies, such as outlining, setting goals, and revising based on feedback.

Within PBL environments, learners negotiate topics, audiences, and genres, strengthening ownership and responsibility for their texts. It is important to highlight that, when using AI tools, autonomy not only includes managing writing strategies, but also regulating technological assistance, which requires deciding when AI functions as scaffolding and when it may replace cognitive effort.

Recent research highlights that explicit instruction in metacognitive writing strategies improves organization, revision quality, and self-efficacy (Han, 2024; Khosravi et al., 2023). These factors are particularly relevant in higher education, where learners are expected to produce independent academic work.

Mota (2025) emphasizes that project-centered methodologies develop higher-order competencies such as autonomous learning and collaboration. Similarly, Fujii (2024) links flexible learning environments with increased motivation and writing proficiency.

In higher education EFL programs, writing autonomy is therefore both a pedagogical goal and an institutional necessity, especially in contexts where AI technologies can either enhance or undermine independent authorship.

### 2.3 AI in academic writing: benefits, risks, and cognitive implications

ChatGPT is used for brainstorming, drafting, editing, and language support (Sallam et al., 2024). Studies indicate improvements in coherence, linguistic complexity, and idea generation (Herbold et al., 2023; Pikhart et al., 2024).

However, concerns include plagiarism, authorship misrepresentation, reduced critical engagement, and algorithmic inaccuracies (Imran and Almusharraf, 2023; Tai et al., 2023). These tensions highlight a fundamental pedagogical dilemma: AI can support the development of writing, but uncritical dependence on it may weaken the metacognitive skills that are necessary for autonomy.

Research consistently raises concerns about overdependence on AI tools in higher education (Jo, 2024). While AI-supported tasks may enhance productivity, excessive reliance can reduce learners' engagement in planning and problem-solving (Sallam et al., 2024).

Consequently, what matters is not whether AI improves text quality, but whether learners are cognitively responsible for the writing process. This distinction is essential when interpreting advances in post-test performance.

## 2.4 AI literacy and responsible use in education

AI literacy involves understanding, evaluating, and ethically integrating AI technologies (Chiu, 2025; Salhab and Aboushi, 2025). Responsible use frameworks emphasize transparency, accountability, privacy, and academic integrity (Smith et al., 2026; Temper et al., 2025).

Educational criteria for AI integration include pedagogical alignment, reliability, and preservation of cognitive engagement (Kalniņa et al., 2024; Madanchian and Taherdoost, 2025).

For EFL learners participating in writing projects, AI management becomes an integral part of autonomy: learners must critically consider AI suggestions, verify information, and preserve authorship. Consequently, responsible AI use is not an external regulation, but an extension of self-regulated learning.

## 2.5 Connecting PBL, writing autonomy, and responsible AI use

Project-based learning fosters autonomy by requiring learners to make sustained decisions during planning, drafting, collaboration, and revision (Khoudri et al., 2023; Mahmud and Islam, 2022). These processes inherently strengthen self-regulation.

Autonomous learners are more likely to use AI strategically rather than dependently (Kalniņa et al., 2024; Mohebbi, 2025). In PBL contexts, authentic problem-solving demands original contributions, making passive AI substitution less viable.

PBL can therefore serve as a scaffolding framework: rather than framing AI as beneficial or harmful, it places it in a structured environment that promotes metacognitive control, collaboration, and reflective decision-making.

This theoretical integration provides the basis for the present study, which explores whether a brief intervention based on problem-based learning can improve writing skills while promoting greater autonomy and more self-regulated patterns of AI use among university learners who are learning English as a foreign language (EFL).

## 3 Methodology

### 3.1 Design

This study used a quasi-experimental, single-group pretest-posttest design to examine the effects of project-based learning (PBL) on EFL university learners' writing proficiency, writing autonomy, and patterns of AI use.

Quantitative data were collected to compare learners' performance and technology-related behaviors before and after the intervention. In the pretest, learners were allowed to use ChatGPT according to their typical writing practices in order to capture baseline performance under authentic conditions. After a one-week PBL intervention, the post-test was administered under controlled conditions in which AI assistance was explicitly prohibited.

The change in AI access requirements was intentional and had a theoretical basis. While the pre-test reflected learners' writing habits (including possible dependence on AI), the post-test was designed to assess independent writing performance, which was used in this study as an indicator of writing autonomy.

Since the study had a single-group design without a control group, causal inferences should be interpreted with caution. The improvements observed in post-test scores may reflect the combined influence of the PBL intervention, increased proficiency with the email writing process, short-term instructional exposure, and testing effects. Therefore, findings are analyzed as associations rather than definitive causal statements.

### 3.2 Participants and procedure

The initial sample included 67 A2-level EFL students from Universidad Central del Ecuador. They were selected through non-probability convenience sampling, which was appropriate for intact classroom groups (Hernández-Sampieri and Mendoza, 2018). Of the 67 learners enrolled at the beginning of the academic year, 56 were present on the day the pre-test was administered. There was a dropout rate due to normal institutional attrition and variability in attendance. At the time of the post-test administration, 44 learners completed both the pre-test and the post-test. Only these paired cases ( $n = 44$ ) were included in the final statistical analysis of paired samples to ensure internal consistency and scores comparison.

This progressive reduction in sample size reflects natural classroom attrition rather than experimental exclusion. However, when interpreting generalized results, the decrease in the number of participants should be taken into account.

Although no formal *a priori* power analysis was conducted, the final sample size of 44 aligns with conventional recommendations for detecting medium effect sizes in paired-sample t-tests at  $\alpha = .05$  (Cohen, 1992). Nevertheless, statistical power limitations should be considered when non-significant findings are interpreted in survey data.

#### 3.2.1 Structure of PBL intervention

To enhance replicability, the one-week PBL intervention followed a structured five-session sequence integrated into



regular class hours (five sessions of approximately 90 min each) (see [Figure 1](#)). Learners worked in groups of four to five members.

Given the compressed timeframe, the intervention aimed to intensively model collaborative writing processes rather than produce long-term developmental change.

### 3.3 Instruments

#### 3.3.1 Pretest and post-test

The writing tasks were modeled on the Cambridge A2 Key email-writing format. The analytic rubric used is based on the principles of criteria based assessment ([Boud and Falchikov, 2007](#)), which stress that assessment of learners should include transparency, alignment to learning outcomes, and a formative purpose. By having performance descriptors, learners can have access to what is expected of them, and they can provide their peers with feedback ([Boud and Falchikov, 2007](#); [Mertens, 2020](#)).

In the pretest, learners wrote an email proposing a new meeting place due to a cancelled reservation (see [Supplementary Appendix A](#)). AI was not restricted. In the post-test, learners wrote an email suggesting alternative travel arrangements due to a train cancellation (see [Supplementary Appendix B](#)), and AI assistance was prohibited.

Both tasks shared the same communication format (informal transactional email) and a similar level of difficulty in order to keep the structure consistent and avoid identical prompts. However, as both tasks shared the same format, part of the improvement may be a consequence of increased familiarity with the email structure and not only the PBL intervention. This limitation is addressed in the interpretation of the results.

#### 3.3.2 Scoring procedure and reliability

The co-authors of this research used a common analytic rubric with a total score of 15 points for assessing the aspects of content, organization, and language (grammar and vocabulary) of all writing samples from the participants in the study. To establish a shared understanding of the analytical rubric descriptors and rubric score levels (each criterion was scored on a scale of 0–5),

calibrating sessions preceded the scoring in which a number of sample writing scripts were jointly assessed and discussed. This last step was to limit subjective variations and to increase reliability among scorers.

After calibration was completed, the three raters were then assigned several writing scripts to rate independently using the analytic rubric described above and previously calibrated prior to the rating process. Due to the absence of double scored writing scripts, inter-rater reliability coefficients were not computed.

##### 3.3.2.1 Surveys

Two structured five-point Likert-scale surveys were created following the survey design guidelines described by [Ponto \(2015\)](#).

##### 3.3.2.2 Learner survey

This instrument measured:

- Frequency of ChatGPT use
- Reliance on AI for grammar, vocabulary, and idea generation
- Self-perceived writing performance
- Writing autonomy indicators (planning, self-checking, confidence writing without AI) (See [Supplementary Appendix C](#))

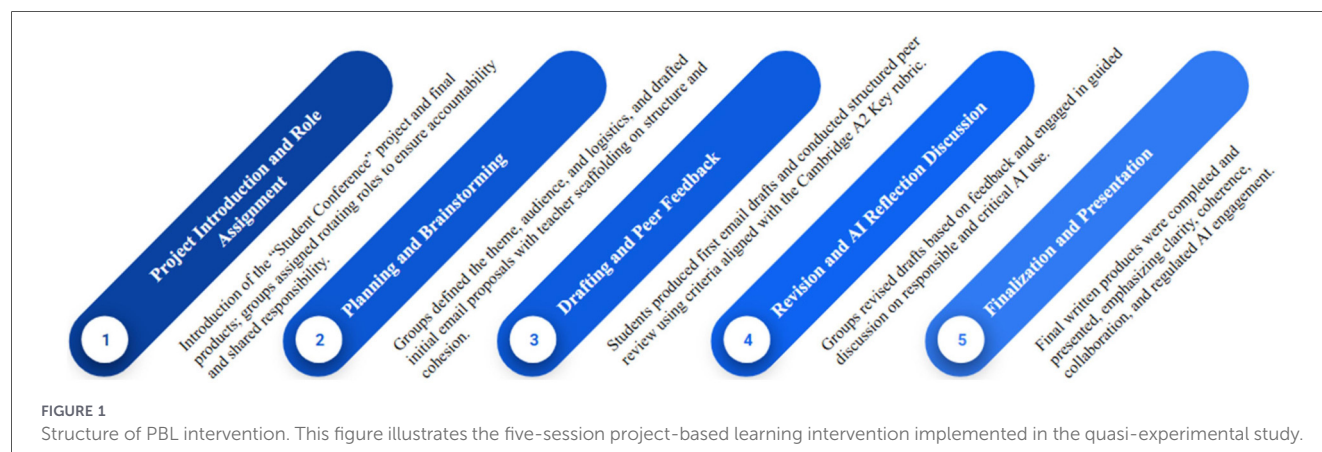
Three TESOL specialists (the authors of this paper) reviewed the instrument for content validity. Cronbach's alpha was .81 (see [Supplementary Appendix D](#)), indicating strong internal consistency ([Nunnally and Bernstein, 1994](#), p. 248–292).

Since autonomy indicators were self-reported, findings related to autonomy in writing reveal perceived metacognitive behavior rather than directly observed behavior.

##### 3.3.2.3 Teacher survey

This survey measured:

- Familiarity with PBL
- Perceptions of learner collaboration
- Task division and problem-solving
- Motivation during PBL



- Observations of learner independence in writing (Supplementary Appendix E)

Cronbach's alpha for this survey was .60 (see Supplementary Appendix F).

Due to the fact that only two people completed the teacher survey, the reliability coefficients are not considered statistically significant and are not interpreted.

### 3.4 Data analysis

Descriptive statistics were calculated for the writing scores from the surveys, plus sample-t test was used to find the statistically significant difference from pretest to posttest writing performance to find out if there was a difference as a result of the intervention. Cohen's *d* for paired samples was calculated to show how much change has happened statistically, but also to show the amount of change independent of statistical significance.

Survey data were analyzed using descriptive statistics and cross tabulations were used to find the change in use of AI, dependency on AI, planning behaviour, and confidence to write without AI.

Drawbacks in the design of the study indicate that the statistical results of this study should be interpreted as indicating short term educational impact, and not as evidence of causality.

### 3.5 Ethical considerations

The Academic Language Institute's academic committee reviewed and approved the study. Participants received information about the study's purpose and procedure and provided written consent. Learners were assured that participation was voluntary, confidentiality would be

maintained, and academic evaluations would not be affected (see Supplementary Appendices G, H).

All instruments underwent expert validation. AI tools were used solely to improve wording and formatting of survey items and rubrics, not for data analysis. These measures ensured compliance with ethical research practices.

## 4 Findings

### 4.1 Writing performance: pre- and post-test outcomes

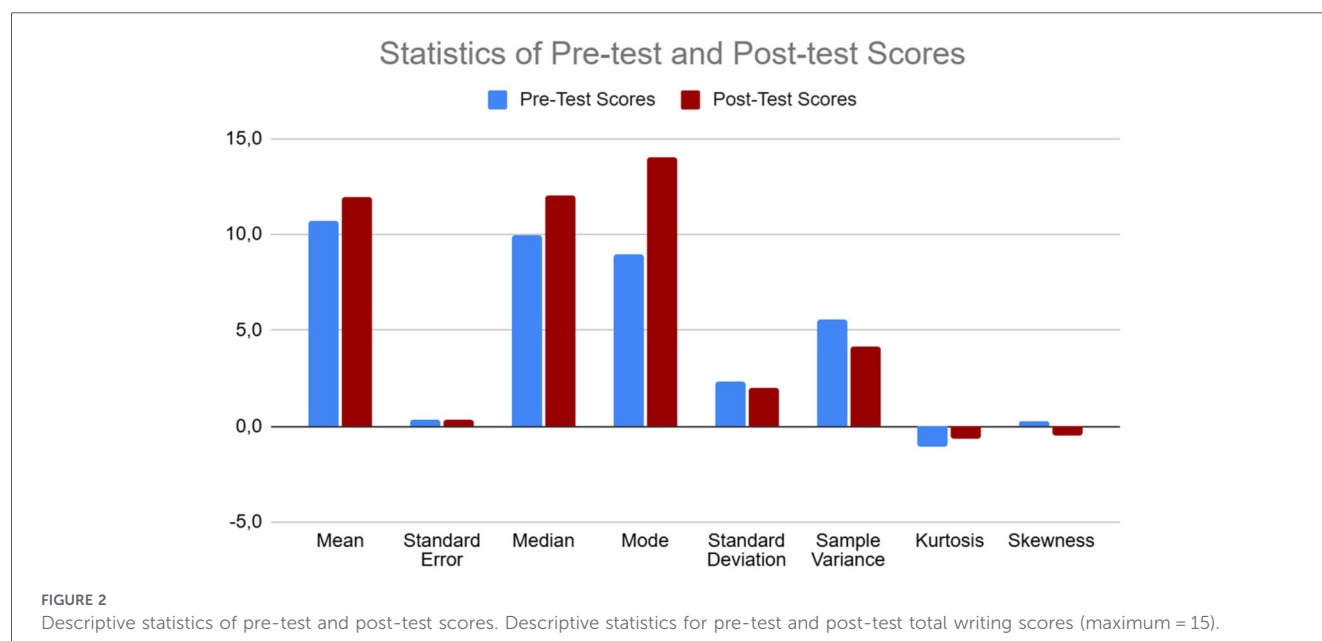
The writing score analysis used matched data from 44 learners who completed both the pretest and post-test. Initially, 67 learners were enrolled, and 56 finished the pretest. However, only 44 were present for both test occasions. Thus, the paired-samples analysis focused on these matched cases ( $n = 44$ ).

The pretest was conducted under conditions that allowed AI use, reflecting the learners' typical writing practices. The post-test was done without AI assistance.

Descriptive statistics showed differences between the two tests. The average writing score rose from 10.8 (SD = 2.37) in the pretest to 12.0 (SD = 2.03) in the post-test. The median increased from 10 to 12, and the mode went from 9 to 14. The range stayed at 8 points for both tests. Variance dropped from 5.6 to 4.1 (see Figure 2).

Score distribution across intervals also changed. The percentage of learners scoring in the 7–8 range dropped from 15.91% in the pretest to 4.55% in the post-test. Meanwhile, the percentage scoring between 13 and 15 increased from 29.55% to 47.73% (see Table I1 in Supplementary Appendix I).

A paired-samples t-test compared pretest and post-test scores. The analysis found a significant difference between the two test occasions,  $t(43) = 4.15$ ,  $p < .001$  (two-tailed), with an effect size of Cohen's  $d = 0.63$ . (see Tables 12, I3 in Supplementary Appendix I).



## 4.2 Learner survey: ChatGPT usage and writing autonomy

The learner survey was completed by the 67 students enrolled at the beginning of the study, prior to attrition. Responses were analyzed using descriptive statistics. Regarding overall ChatGPT use for writing tasks (see [Table 1](#)):

- 7.46% reported “never”
- 22.40% reported “rarely”
- 37.31% reported “sometimes”
- 25.37% reported “often”
- 7.46% reported “always”

For grammar checking and brainstorming, 37.31% reported “sometimes” and 35.82% reported “rarely”. For vocabulary-related tasks, 41.79% reported “sometimes”, while 32.84% reported “rarely” or “never”.

Perceived dependence responses were as follows:

- 19.40% reported no dependence
- 35.82% reported “a little” dependence
- 32.84% reported “occasional” dependence
- 5.97% reported “very” or “total” dependence

Items related to planning behaviors, self-checking strategies, and perceived ability to write without ChatGPT showed that most responses were concentrated in mid-range categories (e.g., “sometimes”, “okay”, or “good”) (See Figures J1, J2 in [Supplementary Appendix J](#)).

Because these measures were self-reported, they reflect participants’ perceptions.

Regarding learners’ views on their writing performance without using ChatGPT, the results show mainly moderate self-assessment. As shown in [Table 2](#), 46.27% of the participants

rated their writing as bad, while 34.33% thought it was okay. A smaller group, 4.47%, felt their performance was good, and 14.93% rated it as very bad. Notably, none of the participants chose the excellent category. Overall, these findings suggest that most learners see their independent writing ability as below average to average, with very few showing high confidence in their performance without AI help.

## 4.3 Teacher perception of PBL

Two instructors completed the teacher questionnaire. Both indicated prior familiarity with project-based instructional approaches. Ratings of classroom dynamics during the intervention reflected positive evaluations of learner motivation and participation.

For “Task Division” and “Problem-Solving”, one instructor selected “Slightly Effective”, while the other selected “Moderately Effective”.

Due to the small number of respondents ( $n = 2$ ), results are presented descriptively (See [Supplementary Appendix K](#) for the results of the teacher perception surveys).

## 5 Discussion

This study looked at changes in EFL university learners’ academic writing and patterns of AI use after a short project-based learning (PBL) intervention. The findings showed a statistically significant increase in writing scores between the pre-test and post-test. While this improvement happened after PBL was implemented, the quasi-experimental design and lack of a control group mean we cannot say the gains are only due to this teaching method. Additionally, the pre-test allowed AI use according to the learners’ usual practices, while the post-test

TABLE 1 Learners’ self-reported writing and ChatGPT usage behaviors.

| Survey items                  | Never    |       | Rarely | Sometimes | Often  | Always | Total       |
|-------------------------------|----------|-------|--------|-----------|--------|--------|-------------|
| Planning Skills               | <i>f</i> |       | 17     | 24        | 20     | 6      | <b>67</b>   |
|                               | %        |       | 25.37% | 35.82%    | 29.85% | 8.96%  | <b>100%</b> |
| Self-Checking                 | <i>f</i> |       | 7      | 18        | 25     | 17     | <b>67</b>   |
|                               | %        |       | 10.45% | 26.87%    | 37.31% | 25.37% | <b>100%</b> |
| ChatGPT Usage Frequency       | <i>f</i> | 5     | 15     | 25        | 17     | 5      | <b>67</b>   |
|                               | %        | 7.46% | 22.4%  | 37.31%    | 25.37% | 7.46%  | <b>100%</b> |
| ChatGPT Grammar Brainstorm    | <i>f</i> | 2     | 24     | 25        | 12     | 4      | <b>67</b>   |
|                               | %        | 2.99% | 35.82% | 37.31%    | 17.91% | 5.97%  | <b>100%</b> |
| Grammar Vocab Without ChatGPT | <i>f</i> |       | 2      | 20        | 28     | 17     | <b>67</b>   |
|                               | %        |       | 2.99%  | 29.85%    | 41.79% | 25.37% | <b>100%</b> |
| ChatGPT Query Improvement     | <i>f</i> | 3     | 21     | 27        | 9      | 7      | <b>67</b>   |
|                               | %        | 4.48% | 31.34% | 40.30%    | 13.43% | 10.45% | <b>100%</b> |
| ChatGPT Ideas in Writing      | <i>f</i> | 1     | 29     | 24        | 8      | 5      | <b>67</b>   |
|                               | %        | 1.49% | 43.28% | 35.82%    | 11.95% | 7.46%  | <b>100%</b> |

*f* = frequency; % = percentage.  $N = 67$ . Response options were Never, Rarely, Sometimes, Often, and Always. Percentages are based on the total number of respondents. Note: Bold values in the Total column indicate the total number of valid responses for each survey item and the corresponding overall percentage (100%).

TABLE 2 Learners' perceived writing performance without ChatGPT.

| Survey item             | Very Bad | Bad    | Okay   | Good   | Excellent | Total       |
|-------------------------|----------|--------|--------|--------|-----------|-------------|
| Writing Without ChatGPT | <i>f</i> | 10     | 31     | 23     | 3         | <b>67</b>   |
|                         | %        | 14.93% | 46.27% | 34.33% | 4.47%     | <b>100%</b> |

*f* = frequency; % = percentage. *N* = 67. Response options were Very Bad, Bad, Okay, Good, and Excellent. Percentages are based on the total number of respondents.

Note: Bold values in the Total column indicate the total number of valid responses for the survey item and the corresponding overall percentage (100%).

required writing independently without AI help. Therefore, the observed improvement reflects performance under different conditions and should be seen as evidence of change over time, rather than clear proof that PBL alone caused the effect.

The shift in score distributions indicates movement from lower performance bands to higher achievement categories. This pattern matches prior research that suggests PBL can aid writing development in language learning contexts. However, because the intervention was short and the assessment format was the same, other factors like greater familiarity with the writing genre, exposure to instruction, or effects from testing may also play a role.

Survey data gave more context about learners' reported AI use. At baseline, learners frequently relied on ChatGPT for grammar and vocabulary help. Although post-intervention survey responses and observations hinted at a trend toward more selective AI use, these findings are based on self-reported data rather than controlled behavior tracking. Thus, they should be taken with caution. The results do not show that AI use was eliminated; these findings should therefore be interpreted as short-term behavioral patterns and emergent indicators of autonomy rather than evidence of sustained developmental change.

The collaborative structure of the PBL cycle might have influenced changes in learner engagement. Activities like peer review, group discussion, and joint problem-solving required learners to articulate and refine ideas together. These processes fit with theories of PBL that highlight active learning and shared thinking. At the same time, teachers noted that learners faced moderate challenges with delegating tasks and solving problems. These findings suggest that while collaboration can boost engagement, there is still a need for structured support for teamwork, especially in brief interventions.

Motivation was a recurring theme in teacher feedback. Instructors observed increased participation and sustained engagement from learners during project tasks. Mergendoller et al. (2017) argue that motivation is essential for skill development in PBL. Although motivational aspects were not measured using standardized scales, qualitative observations support existing literature that indicates task-based instruction can improve engagement in language learning. The link between motivation, writing performance, and reliance on AI needs further study in more controlled settings.

## 6 Limitations

The following methodological limitations should be considered when interpreting the findings.

First, the use of a single-group pretest and posttest design without a comparison group limits the internal validity of this study. Therefore, while it was clear that writing performance improved, we cannot attribute those improvements solely to the PBL intervention. It is also possible that improvements in performance were due to increased familiarity with the assignment format, the short duration of the instructional intervention, or the effects of the repeated testing.

Second, while three trained evaluators utilized a common rubric and participated in a calibration session before scoring, there was only one evaluator for each of the writing scripts, so inter-rater reliability statistics were not computed; this is another methodological limitation.

Additionally, the use of AI was permitted during the pretest but not permitted during the posttest. This creates a systematic problem in the data collected, as the contrast in performance could be a result of adjustment to the AI-restricted circumstances, not the impact of the instructional intervention.

Another limitation is the fact that the intervention lasted for a single week. Writing autonomy and AI literacy often require sustained exposure to develop. Thus, the findings reflect short-term instructional exposure rather than long-term development. Consequently, any references to autonomy development in this study should be understood as emergent indicators inferred from short-term behavioral performance rather than as robust developmental growth.

Furthermore, measures of AI dependence and autonomy were primarily self-reported via survey instruments; there were no objective data collected on participant behaviour, drafts or interaction logs with AI, thus all conclusions regarding responsible use of AI will be considered indicative and not definitive.

Additionally, only two instructors provided surveys to collect teacher data; the Cronbach's alpha (.60) was relatively low; hence, these results are contextual insights as opposed to confirmatory evidence.

Lastly, the various versions of ChatGPT were not controlled for participants. Doing so would have likely influenced their behaviour while utilising the AI tool; hence, no comparison is available to evaluate differences between these two variables or conditions.

## 7 Conclusion

This study explored how a short intervention using project-based learning, or PBL, aimed to analyze the patterns of EFL learners in a university in Ecuador to improve their writing performance without artificial intelligence assistance. One result was that writing scores improved significantly during a period



when there was a restriction on use of AI. In addition to this, the survey conducted after the intervention indicated that learners were making more selective use of AI.

However, the quasi-experimental approach of the study meant that the lack of a control group, small sample size, and reliance on self-reported data means that the results cannot be considered definitive evidence of a causal effect of the PBL intervention.

Future research employing longitudinal designs, control groups, objective AI usage tracking, and multi-rater scoring procedures could strengthen causal inference and generalizability. Also, it may strengthen reliability procedures by incorporating double-scoring and calculating intraclass correlation coefficients. Moreover, comparison studies between project based learning and other teaching methodologies may provide additional information as to the type of learning environments employed with the use of AI.

While the study does not present definitive evidence that AI has a net negative or positive effect on writing, the study suggests that the design of instruction will determine how learners use emerging technologies such as artificial intelligence. PBL may create a context that allows or encourages learners to engage with their assigned writing tasks while also providing opportunities for learners to take ownership of their writing; however, more long-term and controlled research should be conducted to evaluate the sustainability and extent of these findings.

## Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#), further inquiries can be directed to the corresponding author.

## Ethics statement

The studies involving humans were approved by Universidad Espiritu Santo, Unidad de Postgrados en Educación comité de Etica. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

## Author contributions

MA-M: Conceptualization, Software, Visualization, Writing – original draft, Investigation, Resources, Project administration, Writing – review & editing, Formal analysis, Data curation,

Methodology. MG-J: Formal analysis, Visualization, Writing – original draft, Data curation, Project administration, Methodology, Software, Investigation, Resources, Writing – review & editing, Conceptualization. FS-O: Project administration, Formal analysis, Data curation, Methodology, Visualization, Software, Conceptualization, Writing – review & editing, Investigation, Resources, Writing – original draft. AC-F: Writing – review & editing, Funding acquisition, Supervision, Software, Validation.

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## Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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## Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feduc.2026.1757812/full#supplementary-material>

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